Causal Analysis Report

bpic18 Parcel document.xes.gz-imf

Decision points: 243861

Choice sets: 8

Transitions fitted: 6

Significant (p<0.05): 0

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Step 1 — Association test (logistic regression + permutation)
H□: choice □ util | choice_set.   reject_H0=True → util predicts routing.

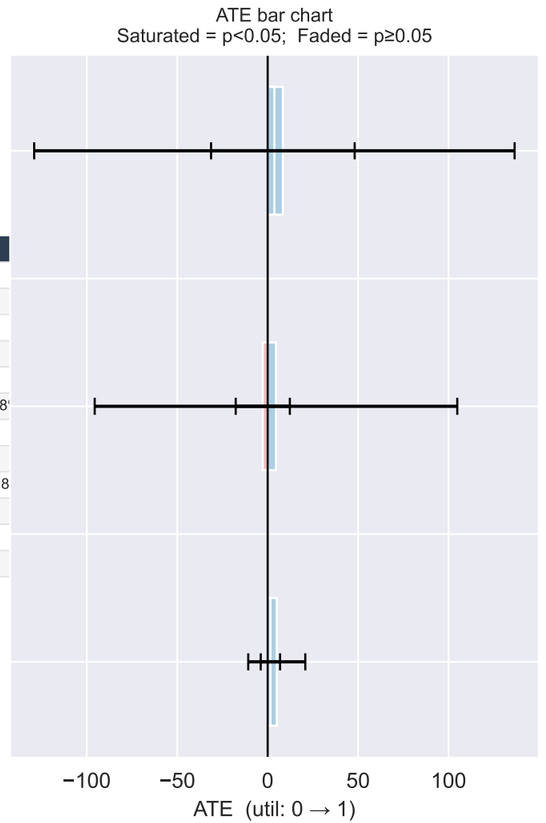
(no data)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

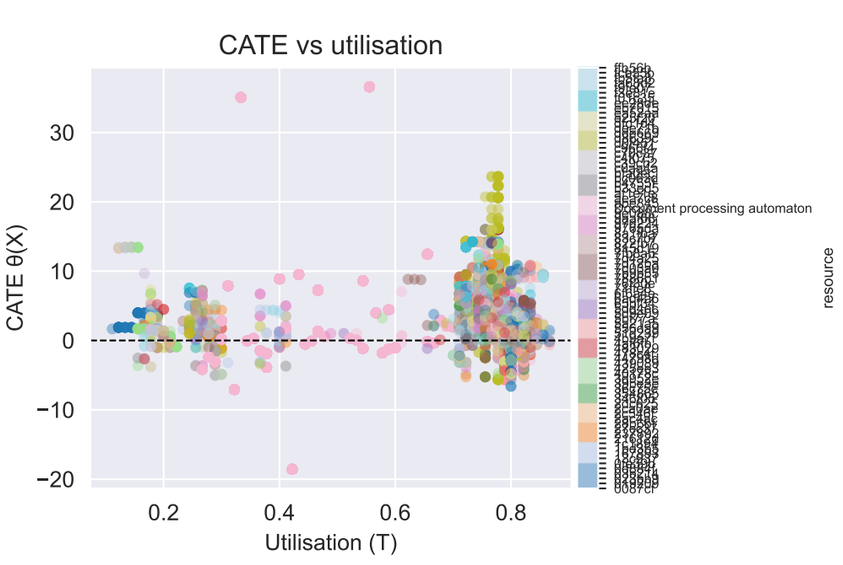
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2)	('τ_t1', 'begin editing_t2')	14551	2	—	—	—	nan	nan	—	nan	—	too few observations
(3, 4)	('τ_t3', 'discard_t4')	3	4	—	—	—	nan	nan	—	nan	—	too few observations
(5, 6)	('τ_t5', 'insert document_t6')	4537	6	—	—	—	nan	nan	—	nan	—	too few observations
(7, 8, 9)	('τ_t7', 'take original document_t8', 'remove document_t9')	5	8, 9	—	—	—	nan	nan	—	nan	—	too few observations
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	11	check_t11	finish editing_t12, save_t13	2.425258	0.250194	[1.9349, 2.9156]	0.0	True	θ=+2.4253: 1-unit ↑ util ↑ P(t11/check_t11) by 242.5%
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	12	finish editing_t12	check_t11, save_t13	-0.104653	0.475452	[-1.0365, 0.8272]	0.8258	False	θ=-0.1047: 1-unit ↑ util ↓ P(t12/finish editing_t12) by 10.5%
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	13	save_t13	check_t11, finish editing_t12	10.717898	0.880238	[8.9926, 12.4432]	0.0	True	θ=+10.7179: 1-unit ↑ util ↑ P(t13/save_t13) by 1071.8%
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	11	check_t11	finish editing_t12, save_t13	1.751108	0.062606	[1.6284, 1.8738]	0.0	True	θ=+1.7511: 1-unit ↑ util ↑ P(t11/check_t11) by 175.1%
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	12	finish editing_t12	check_t11, save_t13	5.577478	0.528277	[4.5421, 6.6129]	0.0	True	θ=+5.5775: 1-unit ↑ util ↑ P(t12/finish editing_t12) by 557.7%
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	13	save_t13	check_t11, finish editing_t12	2.988307	0.612141	[1.7885, 4.1881]	0.0	True	θ=+2.9883: 1-unit ↑ util ↑ P(t13/save_t13) by 298.8%
(14, 15, 16)	('τ_t14', 'τ_t15', 'change department_t16')	5	16	—	—	—	nan	nan	—	nan	—	too few observations
(15, 16)		0		—	—	—	nan	nan	—	nan	—	too few observations

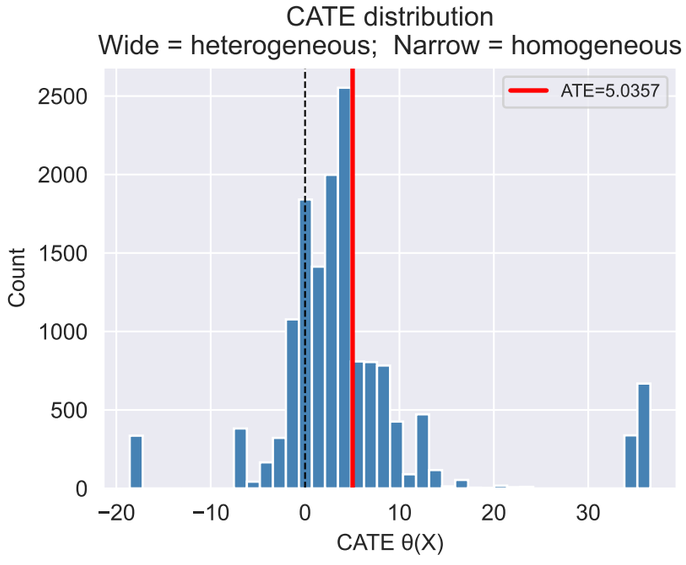
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

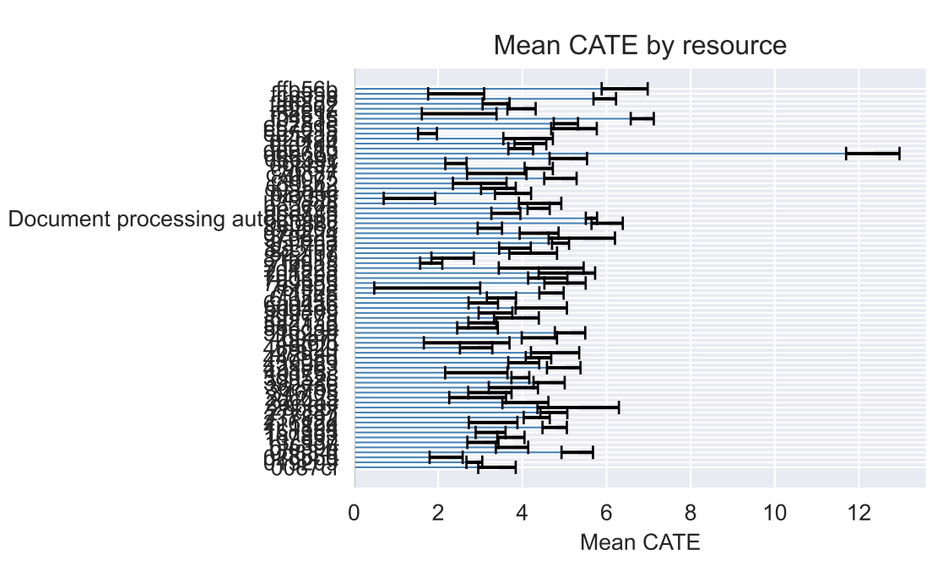
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(1, 2)	('τ_t1', 'begin editing_t2')	14551	2	—	—	—	nan	—	nan	—	too few observations
(3, 4)	('τ_t3', 'discard_t4')	3	4	—	—	—	nan	—	nan	—	too few observations
(5, 6)	('τ_t5', 'insert document_t6')	4537	6	—	—	—	nan	—	nan	—	too few observations
(7, 8, 9)	('τ_t7', 'take original document_t8', 'remove document_t9')	5	8, 9	—	—	—	nan	—	nan	—	too few observations
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	11	check_t11	finish editing_t12, save_t13	5.035715	[-10.7354, 20.8069]	0.5314	False	ATE=+5.0357: 0→1 util ↑ P(t11/check_t11) by 503.6%
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	12	finish editing_t12	check_t11, save_t13	-2.708052	[-17.6734, 12.2573]	0.7228	False	ATE=-2.7081: 0→1 util ↓ P(t12/finish editing_t12) by 270.8
(10, 11, 12, 13)	('τ_t10', 'check_t11', 'finish editing_t12', 'save_t13')	14749	11, 12, 13	13	save_t13	check_t11, finish editing_t12	8.396045	[-31.2990, 48.0911]	0.6785	False	ATE=+8.3960: 0→1 util ↑ P(t13/save_t13) by 839.6%
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	11	check_t11	finish editing_t12, save_t13	1.511254	[-3.8429, 6.8654]	0.5801	False	ATE=+1.5113: 0→1 util ↑ P(t11/check_t11) by 151.1%
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	12	finish editing_t12	check_t11, save_t13	4.597609	[-95.6411, 104.8363]	0.9284	False	ATE=+4.5976: 0→1 util ↑ P(t12/finish editing_t12) by 459.8
(11, 12, 13)	('check_t11', 'finish editing_t12', 'save_t13')	77780	11, 12, 13	13	save_t13	check_t11, finish editing_t12	3.713683	[-129.1111, 136.5385]	0.9563	False	ATE=+3.7137: 0→1 util ↑ P(t13/save_t13) by 371.4%
(14, 15, 16)	('τ_t14', 'τ_t15', 'change department_t16')	5	16	—	—	—	nan	—	nan	—	too few observations
(15, 16)		0		—	—	—	nan	—	nan	—	too few observations

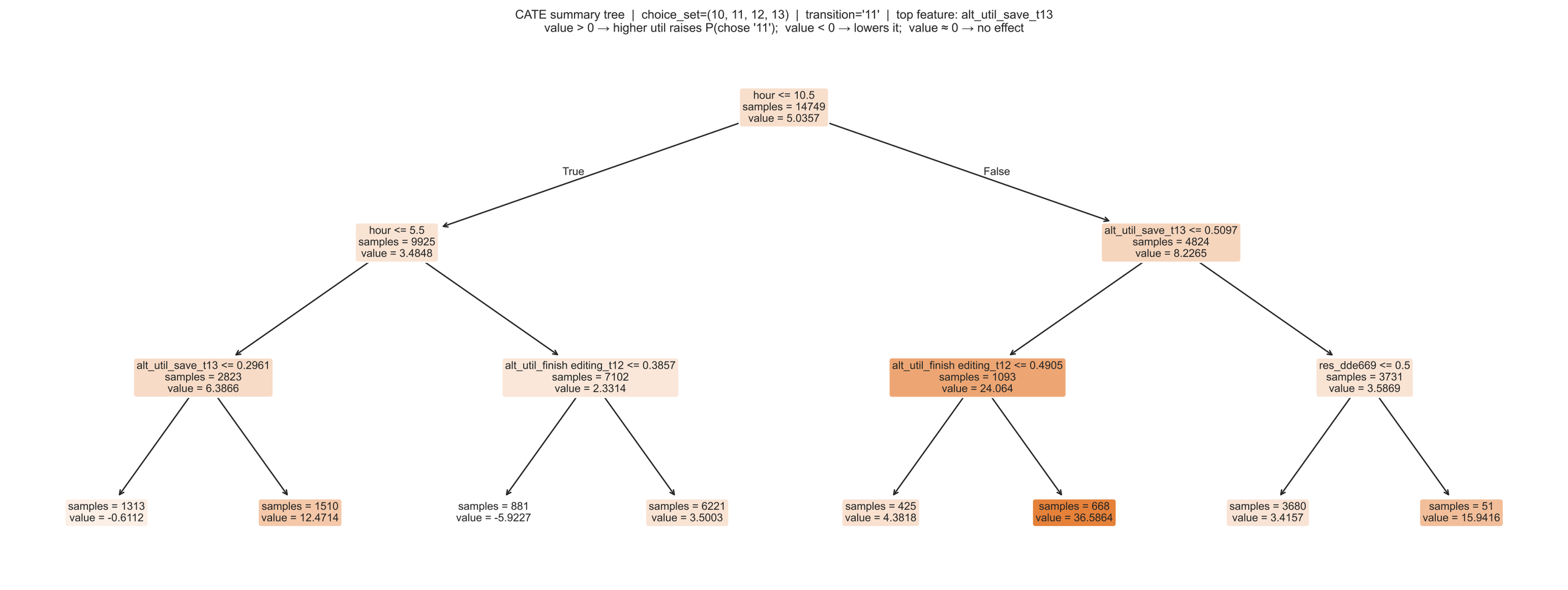


CATE — choice_set=(10, 11, 12, 13) | transition='11' | n=14749 | ATE=5.0357 std=9.8980

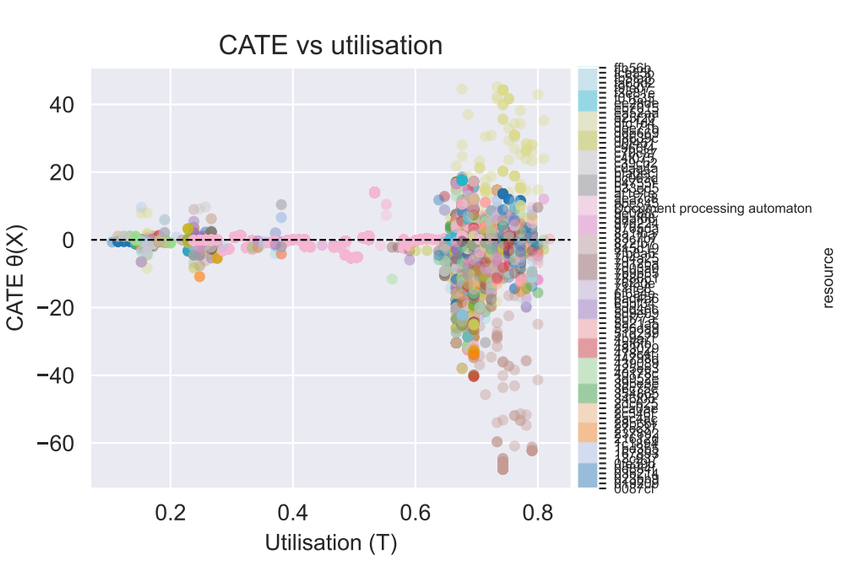


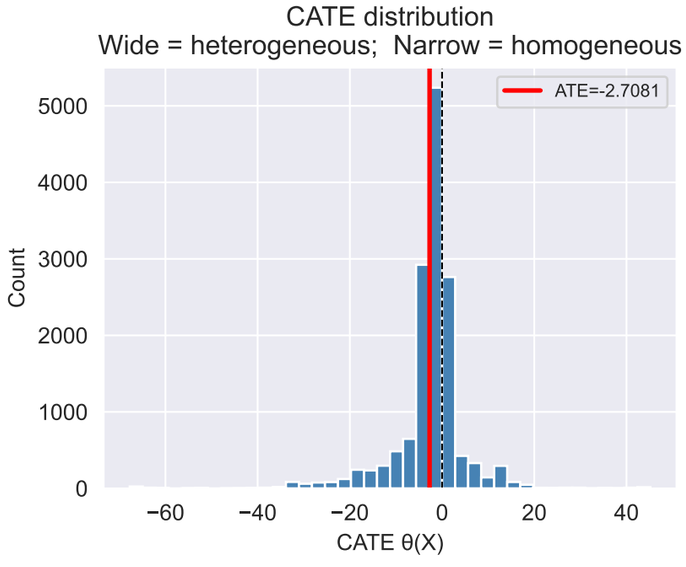


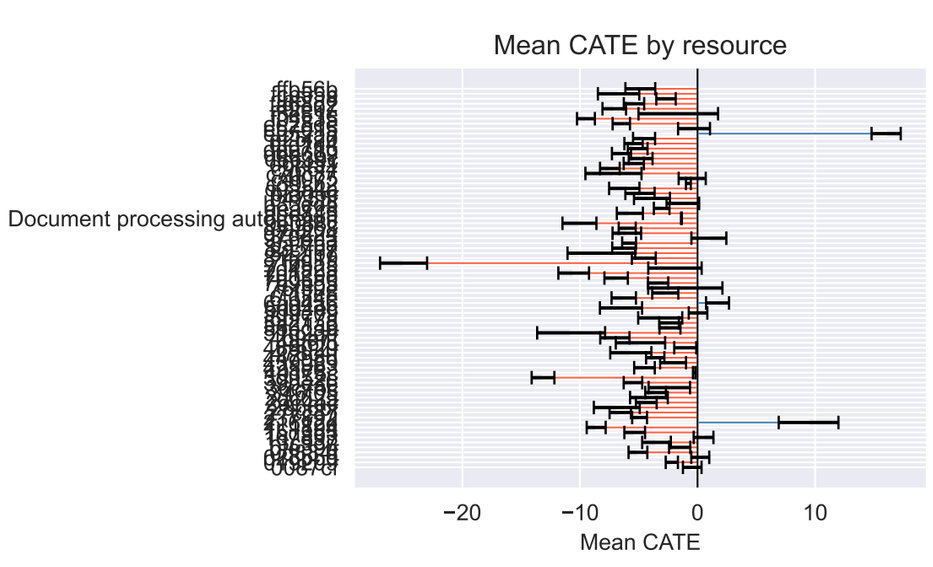


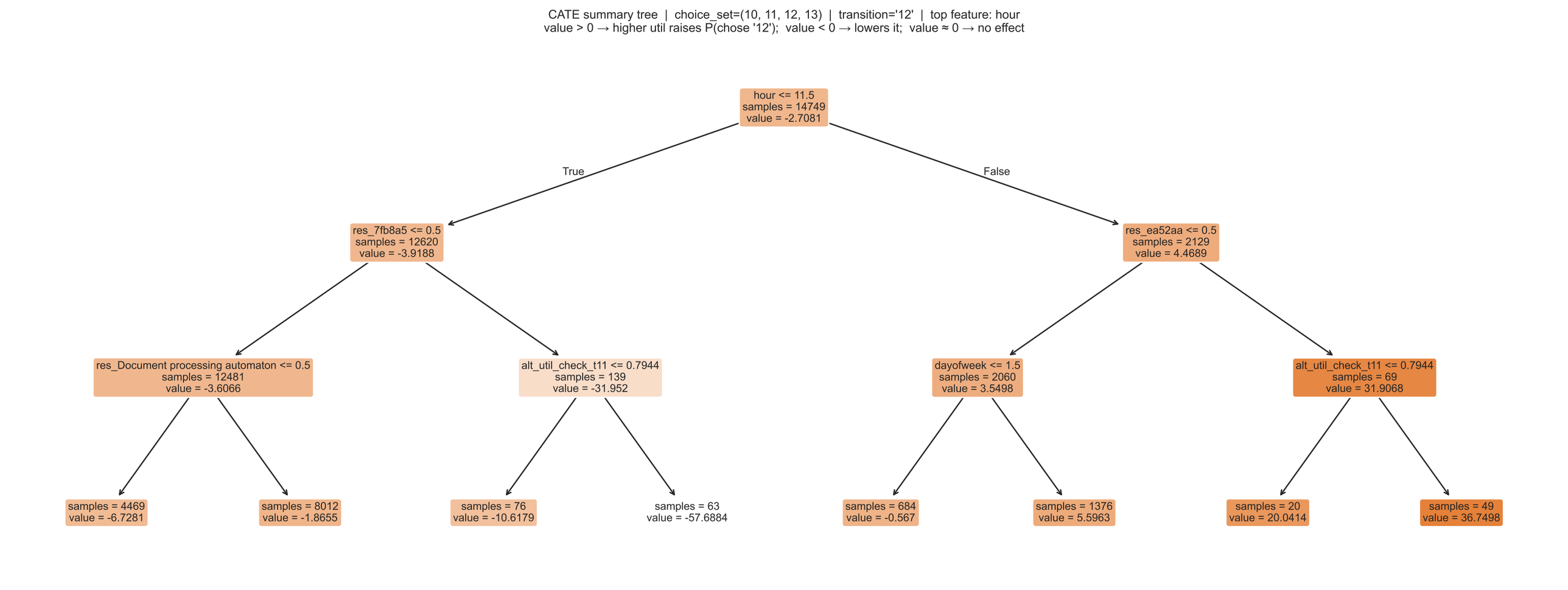

CATE summary tree | choice_set=(10, 11, 12, 13) | transition='11' | top feature: alt_util_save_t13
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 10.5
samples = 14749
value = 5.0357
True
False
hour <= 5.5
samples = 9925
value = 3.4848
alt_util_save_t13 <= 0.5097
samples = 4824
value = 8.2265
alt_util_save_t13 <= 0.2961
samples = 2823
value = 6.3866
alt_util_finish editing_t12 <= 0.3857
samples = 7102
value = 2.3314
alt_util_finish editing_t12 <= 0.4905
samples = 1093
value = 24.064
res_dde669 <= 0.5
samples = 3731
value = 3.5869
samples = 1313
value = -0.6112
samples = 1510
value = 12.4714
samples = 881
value = -5.9227
samples = 6221
value = 3.5003
samples = 425
value = 4.3818
samples = 668
value = 36.5864
samples = 3680
value = 3.4157
samples = 51
value = 15.9416

CATE — choice_set=(10, 11, 12, 13) | transition='12' | n=14749 | ATE=-2.7081 std=8.2283

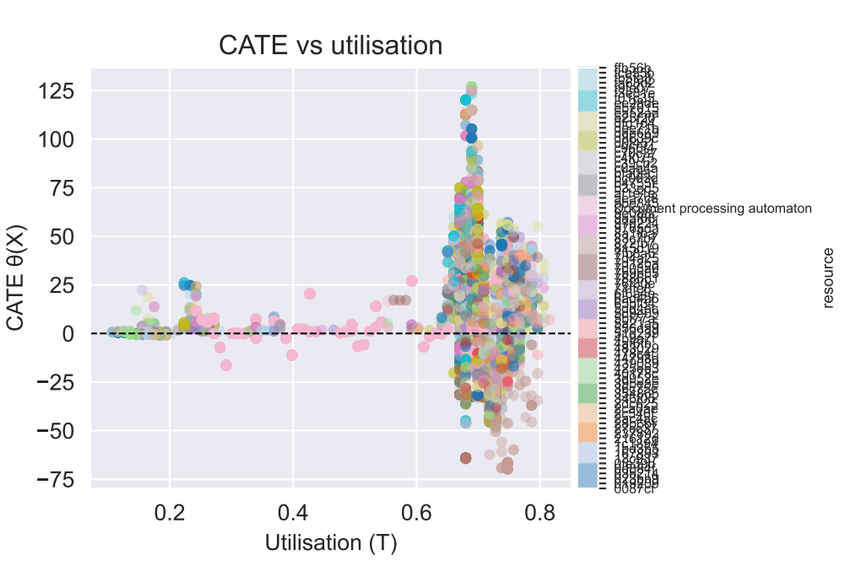


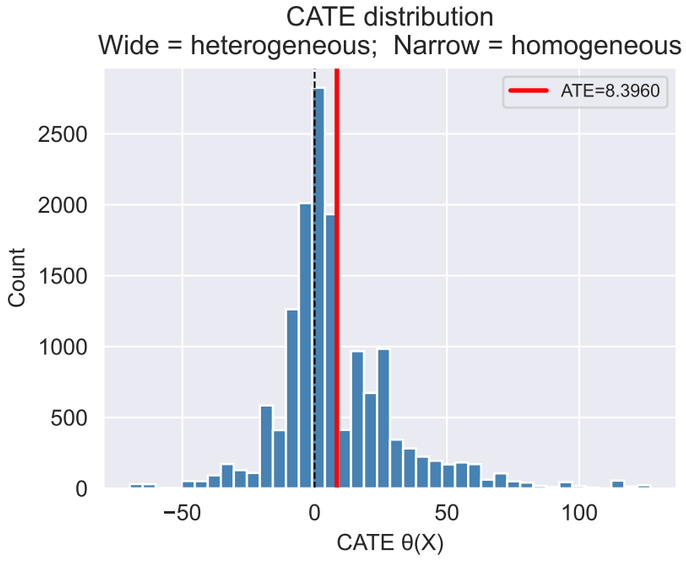


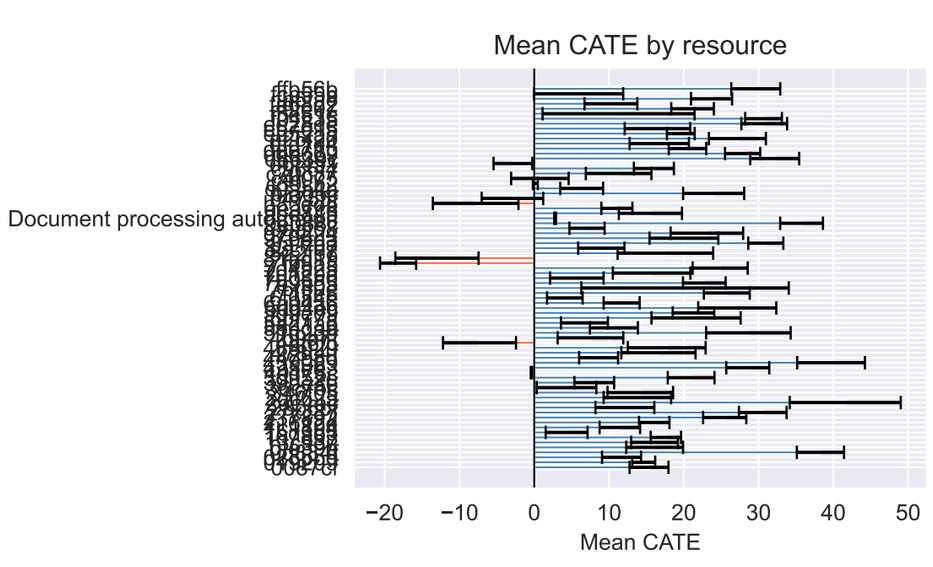


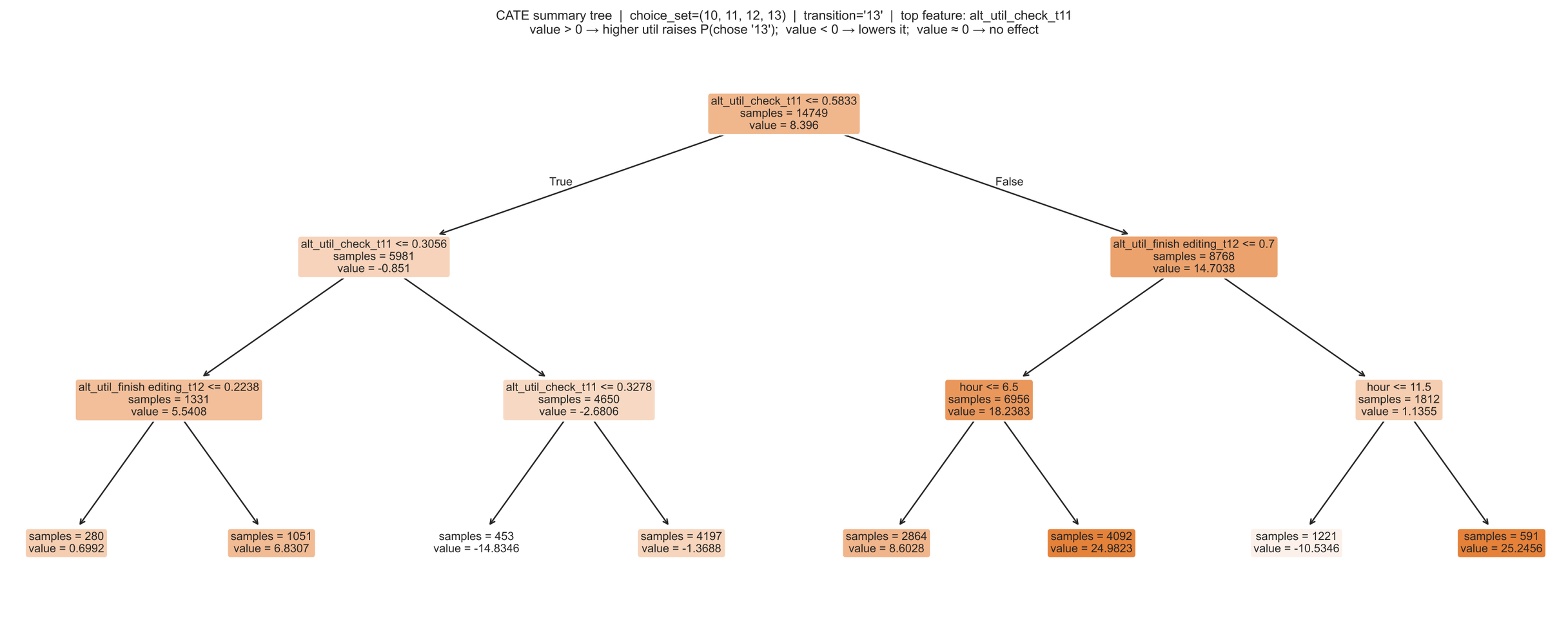

CATE summary tree | choice_set=(10, 11, 12, 13) | transition='12' | top feature: hour
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 11.5
samples = 14749
value = -2.7081
True
False
res_7fb8a5 <= 0.5
samples = 12620
value = -3.9188
res_ea52aa <= 0.5
samples = 2129
value = 4.4689
res_Document processing automaton <= 0.5
samples = 12481
value = -3.6066
alt_util_check_t11 <= 0.7944
samples = 139
value = -31.952
dayofweek <= 1.5
samples = 2060
value = 3.5498
alt_util_check_t11 <= 0.7944
samples = 69
value = 31.9068
samples = 4469
value = -6.7281
samples = 8012
value = -1.8655
samples = 76
value = -10.6179
samples = 63
value = -57.6884
samples = 684
value = -0.567
samples = 1376
value = 5.5963
samples = 20
value = 20.0414
samples = 49
value = 36.7498

CATE — choice_set=(10, 11, 12, 13) | transition='13' | n=14749 | ATE=8.3960 std=23.4592

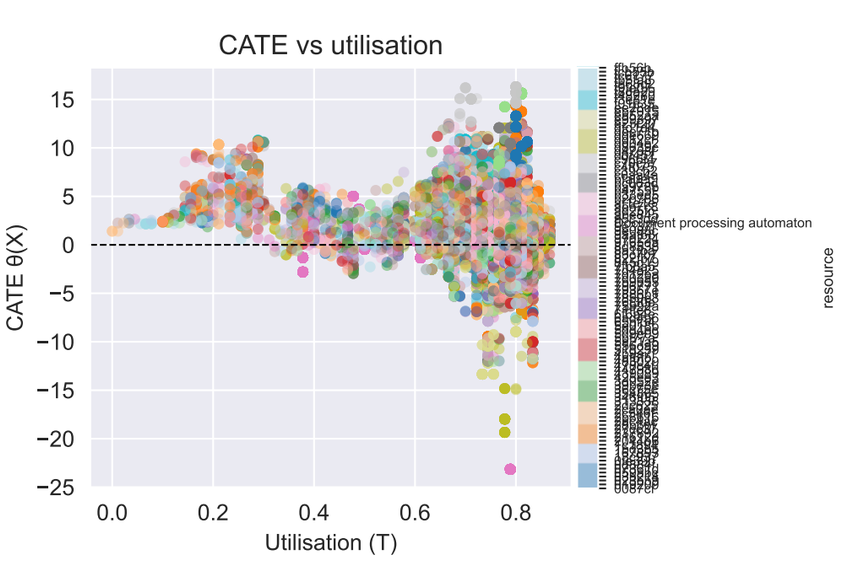


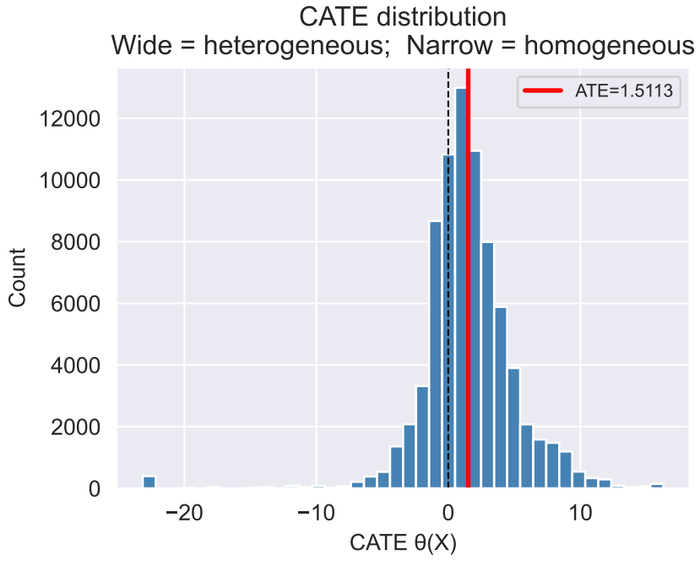


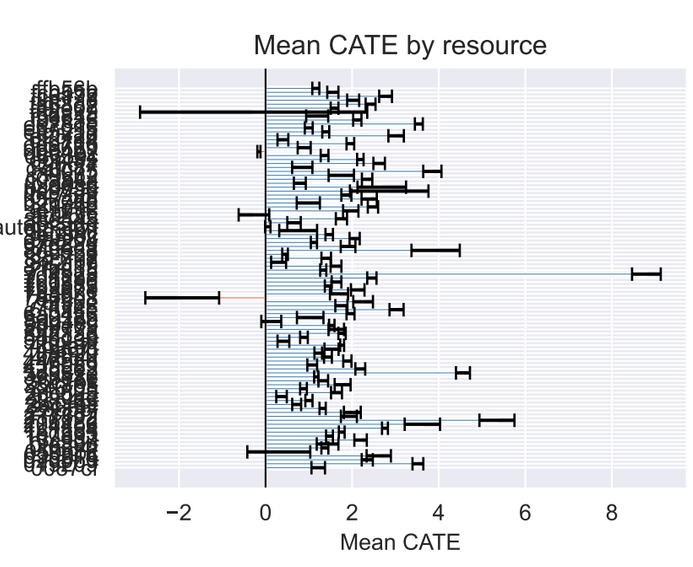


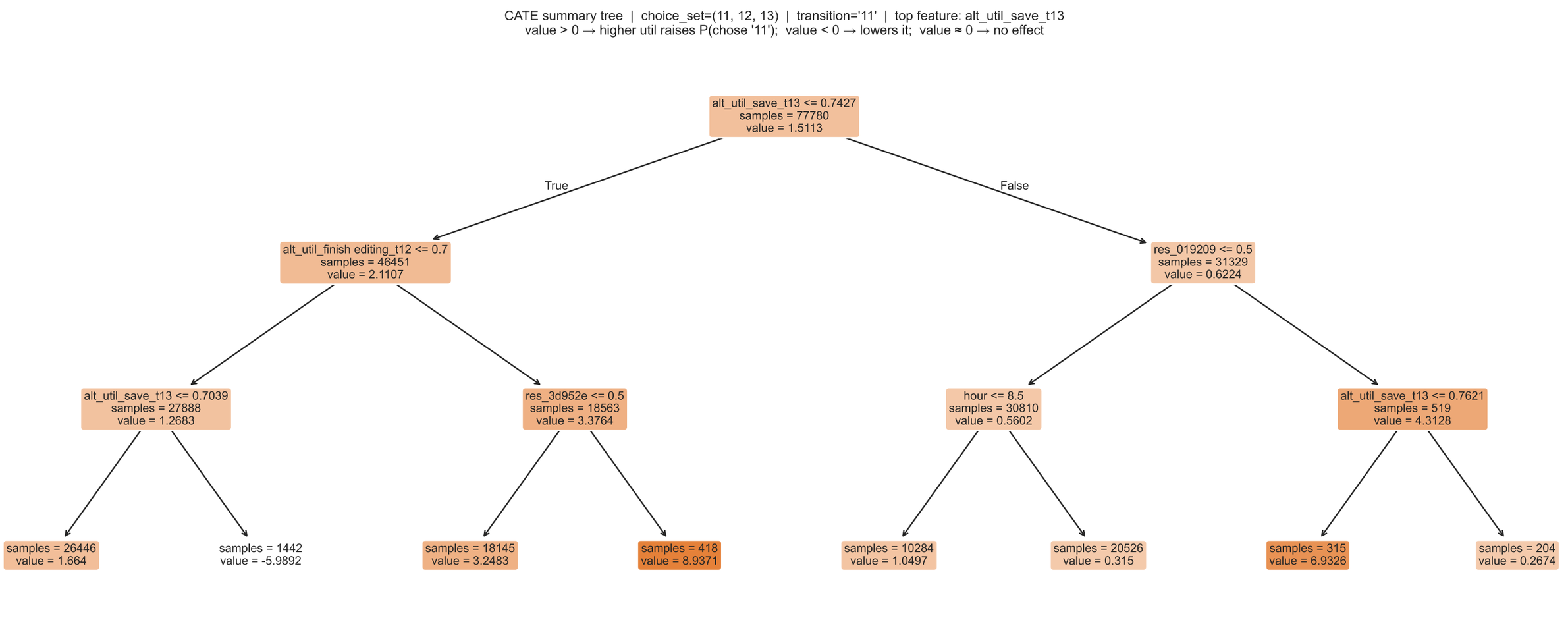

CATE summary tree | choice_set=(10, 11, 12, 13) | transition='13' | top feature: alt_util_check_t11
value > 0 → higher util raises P(chose '13'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_check_t11 <= 0.5833
samples = 14749
value = 8.396
True
False
alt_util_check_t11 <= 0.3056
samples = 5981
value = -0.851
alt_util_finish editing_t12 <= 0.7
samples = 8768
value = 14.7038
alt_util_finish editing_t12 <= 0.2238
samples = 1331
value = 5.5408
alt_util_check_t11 <= 0.3278
samples = 4650
value = -2.6806
hour <= 6.5
samples = 6956
value = 18.2383
hour <= 11.5
samples = 1812
value = 1.1355
samples = 280
value = 0.6992
samples = 1051
value = 6.8307
samples = 453
value = -14.8346
samples = 4197
value = -1.3688
samples = 2864
value = 8.6028
samples = 4092
value = 24.9823
samples = 1221
value = -10.5346
samples = 591
value = 25.2456

CATE — choice_set=(11, 12, 13) | transition='11' | n=77780 | ATE=1.5113 std=3.6720

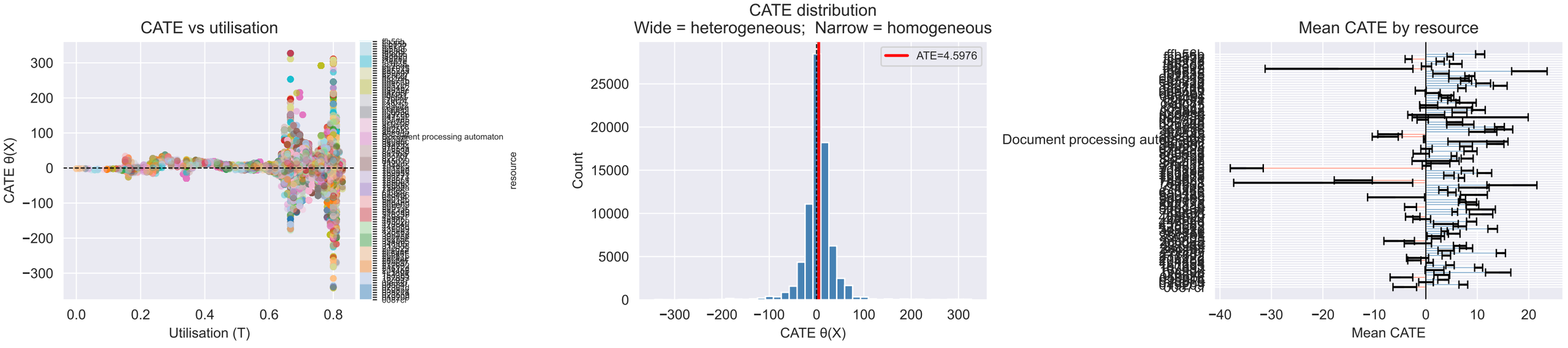


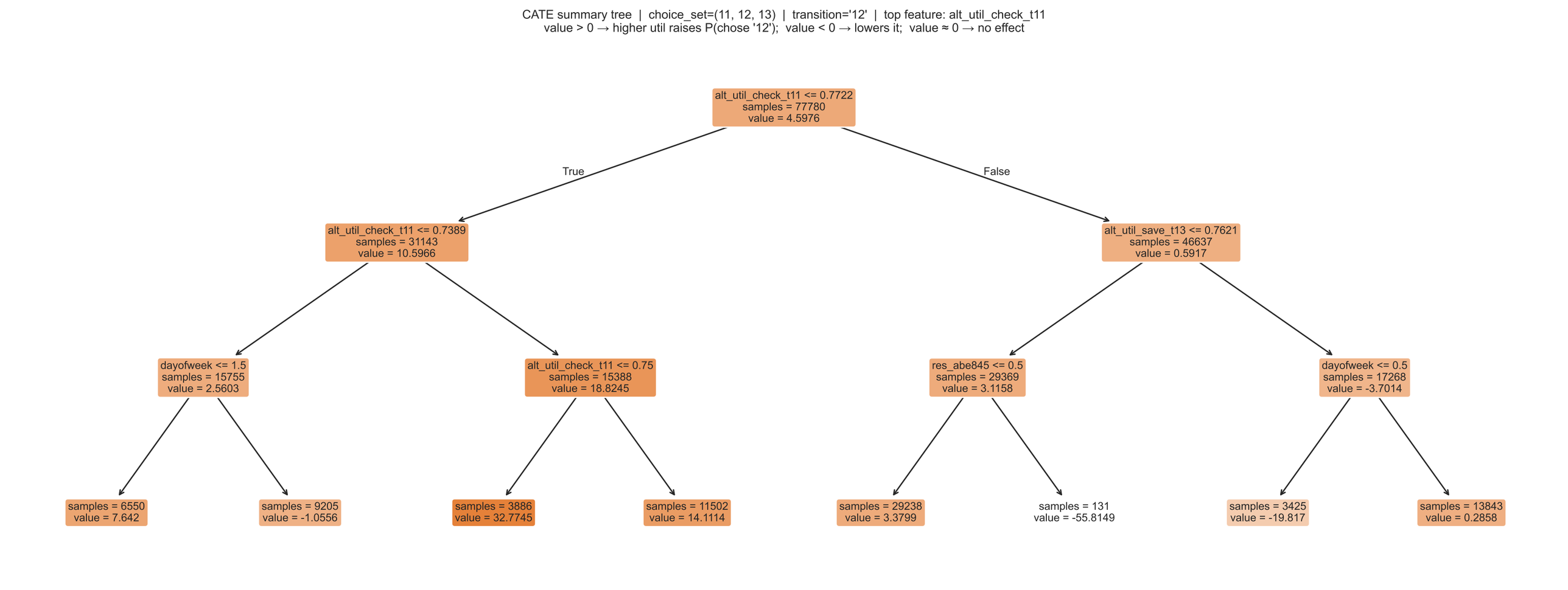




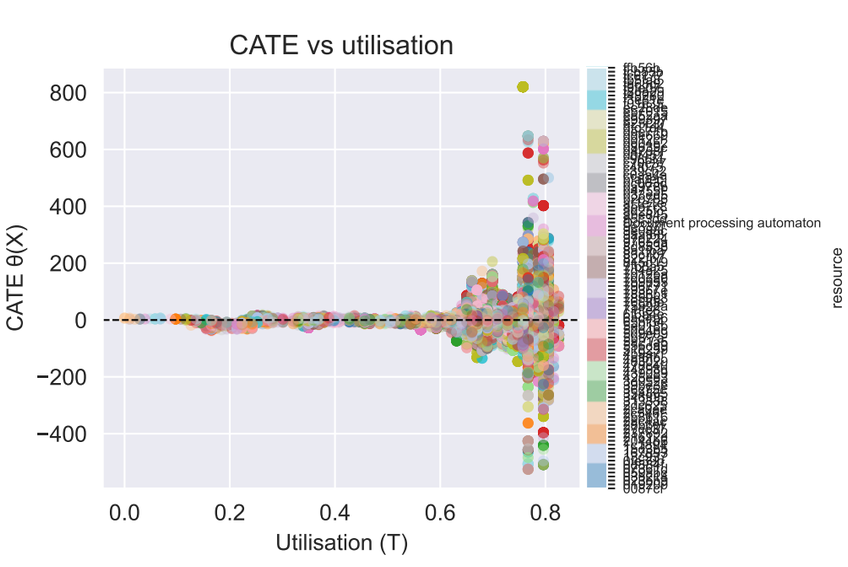

CATE summary tree | choice_set=(11, 12, 13) | transition='11' | top feature: alt_util_save_t13
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_save_t13 <= 0.7427
samples = 77780
value = 1.5113
True
False
alt_util_finish editing_t12 <= 0.7
samples = 46451
value = 2.1107
res_019209 <= 0.5
samples = 31329
value = 0.6224
alt_util_save_t13 <= 0.7039
samples = 27888
value = 1.2683
res_3d952e <= 0.5
samples = 18563
value = 3.3764
hour <= 8.5
samples = 30810
value = 0.5602
alt_util_save_t13 <= 0.7621
samples = 519
value = 4.3128
samples = 26446
value = 1.664
samples = 1442
value = -5.9892
samples = 18145
value = 3.2483
samples = 418
value = 8.9371
samples = 10284
value = 1.0497
samples = 20526
value = 0.315
samples = 315
value = 6.9326
samples = 204
value = 0.2674

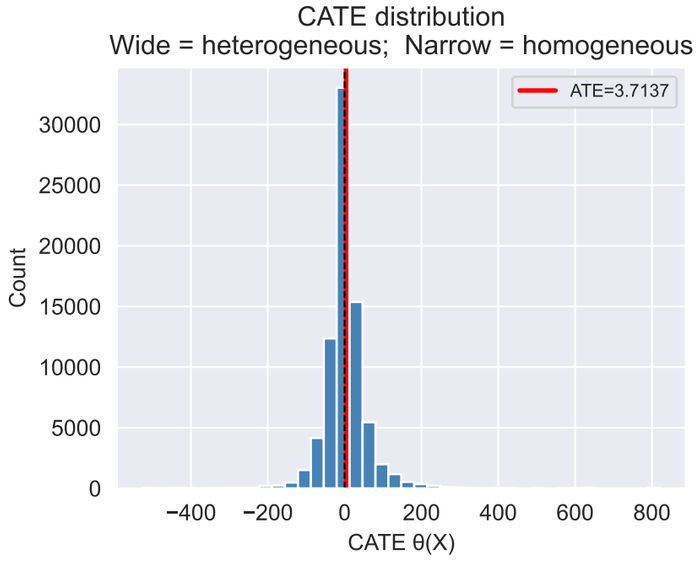
CATE — choice_set=(11, 12, 13) | transition='12' | n=77780 | ATE=4.5976 std=37.2206

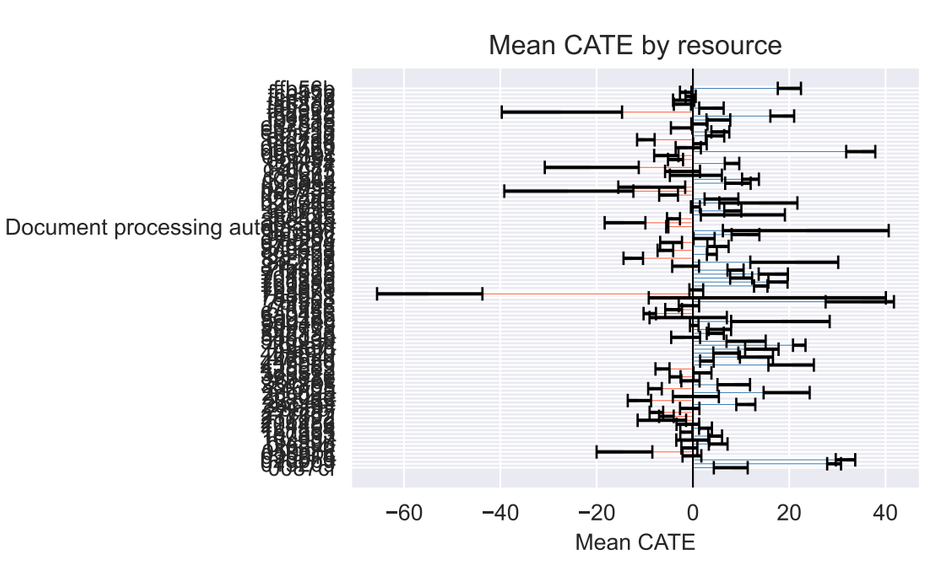


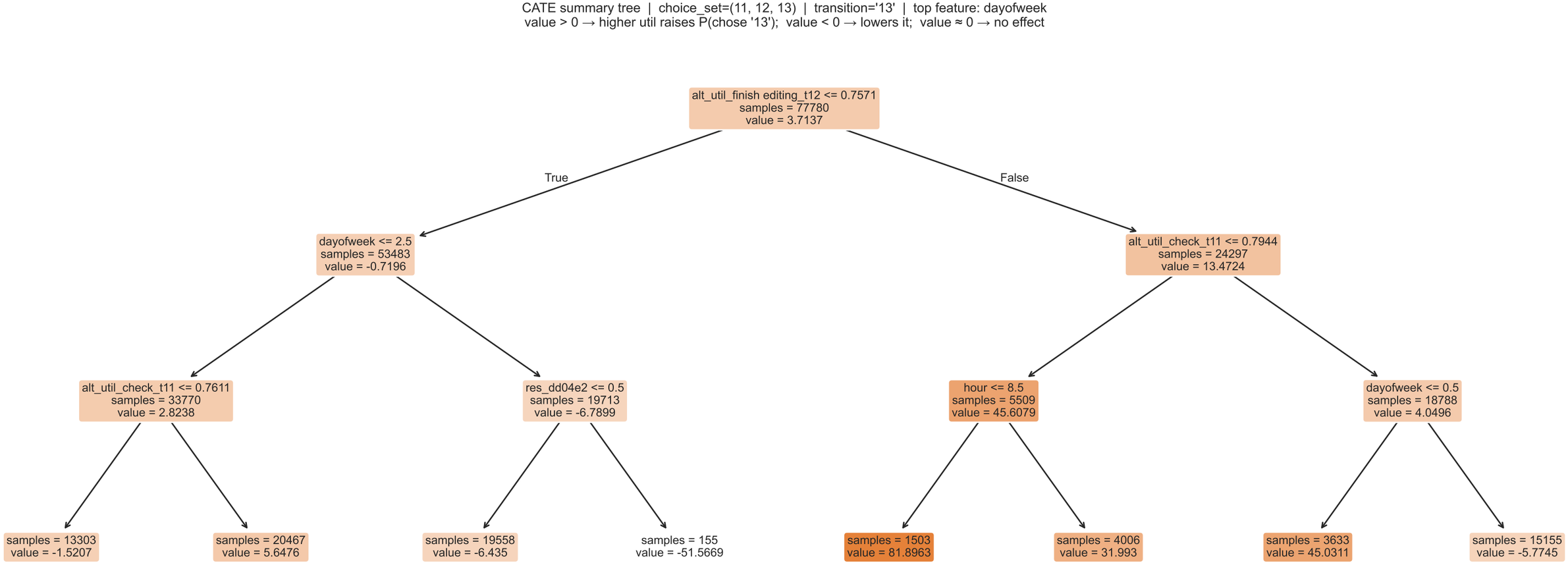

CATE summary tree | choice_set=(11, 12, 13) | transition='12' | top feature: alt_util_check_t11
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_check_t11 <= 0.7722
samples = 77780
value = 4.5976
True
False
alt_util_check_t11 <= 0.7389
samples = 31143
value = 10.5966
alt_util_save_t13 <= 0.7621
samples = 46637
value = 0.5917
dayofweek <= 1.5
samples = 15755
value = 2.5603
alt_util_check_t11 <= 0.75
samples = 15388
value = 18.8245
res_abe845 <= 0.5
samples = 29369
value = 3.1158
dayofweek <= 0.5
samples = 17268
value = -3.7014
samples = 6550
value = 7.642
samples = 9205
value = -1.0556
samples = 3886
value = 32.7745
samples = 11502
value = 14.1114
samples = 29238
value = 3.3799
samples = 131
value = -55.8149
samples = 3425
value = -19.817
samples = 13843
value = 0.2858

CATE — choice_set=(11, 12, 13) | transition='13' | n=77780 | ATE=3.7137 std=64.3433








CATE summary tree | choice_set=(11, 12, 13) | transition='13' | top feature: dayofweek
value > 0 → higher util raises P(chose '13'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t12 <= 0.7571
samples = 77780
value = 3.7137
True
False
dayofweek <= 2.5
samples = 53483
value = -0.7196
alt_util_check_t11 <= 0.7944
samples = 24297
value = 13.4724
alt_util_check_t11 <= 0.7611
samples = 33770
value = 2.8238
res_dd04e2 <= 0.5
samples = 19713
value = -6.7899
hour <= 8.5
samples = 5509
value = 45.6079
dayofweek <= 0.5
samples = 18788
value = 4.0496
samples = 13303
value = -1.5207
samples = 20467
value = 5.6476
samples = 19558
value = -6.435
samples = 155
value = -51.5669
samples = 1503
value = 81.8963
samples = 4006
value = 31.993
samples = 3633
value = 45.0311
samples = 15155
value = -5.7745

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CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(10, 11, 12, 13)	11	14749	5.035715	9.898018	-18.578938	0.431094	3.233189	6.58309	36.586381	81.2	18.8	alt_util_save_t13
(10, 11, 12, 13)	12	14749	-2.708052	8.228338	-67.751126	-5.090375	-1.061532	0.062451	45.18894	28.2	71.8	hour
(10, 11, 12, 13)	13	14749	8.396045	23.459213	-69.901282	-5.760175	2.376336	18.208494	126.911087	60.7	39.3	alt_util_check_t11
(11, 12, 13)	11	77780	1.511254	3.672013	-23.157706	-0.228002	1.302466	3.188507	16.301401	72.0	28.0	alt_util_save_t13
(11, 12, 13)	12	77780	4.597609	37.220647	-342.333512	-7.432135	4.520563	17.468397	327.284515	61.0	39.0	alt_util_check_t11
(11, 12, 13)	13	77780	3.713683	64.343345	-525.109534	-20.223604	0.218381	22.783743	820.358557	50.3	49.7	dayofweek

Key Assumptions & Interpretation Notes

Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md